Sensitivity via Omitted Variable Bias

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Learning sensitivity analysis

Sensitivity analysis is vital to practical causal inference:

- emphasizes the role of assumptions and encourages careful thought about them
- gives you a practical way to work under imperfect identification
- gives you a lens through which to view existing work beyond just complaining about non-zero confounding

Your job

These slides would give you a starting point, but on the course website you will find materials **you should read**, including:

- *From "Is it unconfounded?" to "How much confounding would it take?" The sensitivity-based approach to observational studies*
- *Making Sense of Sensitivity: Extending omitted variable bias*
- *sensemakr: Sensitivity Analysis Tools for OLS in R and Stata*
- *Wildfire Exposure Increases Pro-Climate Political Behaviors*
- Demos at http://carloscinelli.com/sensemakr/articles/

Effects of Violence on Attitudes: Angry or Weary?

If you have seen what we went through you will understand. They came around 6pm on a Sunday...they started looting and burning everything. I am so angry...The only solution is to fight. There is no other way.

If I built a school in Darfur I will call it the "School of Peace", I love peace, and I love to see development, stability, and peace. The peace I long for is not just peace for various tribes, it is a peace for all Sudan...

What is the average effect of exposure to violence on attitudes toward peace (more "angry" or more "weary"), among Darfurian refugees in eastern Chad?

Effects of Violence on Attitudes: Angry or Weary?

Central ID claim: Targeting of violence in 2003-2004 was based only on village and gender but otherwise so indiscriminate as to be independent of (potential) attitudes within village-gender strata".

- Purpose of violence was generally to remove the population rather than kill those likely to be rebels
- Bombing from the air: extremely crude, difficult to target, and no information
- Attacks by *Janjaweed* militia: also highly indiscriminate, had only visual characteristics on which to target (notably, *female*)

What to estimate

Strategy: SOO, conditioning on village and gender

Setup:

- Outcome: *PeaceIndex*, a 0-to-1 scale of pro-peace attitudes
- Treatment: *DirectHarm*, binary measure of being directly injured

Estimation: simple parametric model

$$PeaceIndex_i = \alpha DirectHarm_i + \beta^\top Village_i + \phi female_i + \epsilon_i$$

Finds that those directly harmed are on average 9.7 percentage points more pro-peace (SE=2.3)

But what if the ID strategy is imperfect?

Sensitivity analysis

Numerous sensitivity analyses can be found, but not yet widely used.

In economics,

- Oster (2014) reports that most non-experimental empirical papers utilized only informal robustness tests based on coefficient stability in the face of adding or dropping covariates.

In political science,

- Only 6% of papers involving a causal ID strategy other than experiments conducted a formal analysis of sensitivity to unobserved confounders (covering APSR, AJPS, JOP in 2017 and in 2019).

Questions we'd like sensitivity analysis to help with:

1. What if somebody names a potential confounder I don't have in my data? How "bad" would it have to be to undo my answer?
2. What about multiple confounders – perhaps those I can't name – acting together, even non-linearly?
3. How can we easily communicate overall robustness of a regression results to unobserved confounding?
4. How can we assess whether the sorts of confounders that would alter our conclusions might exist in our case?
5. What if we have a weak identification strategy – can we say anything about what effect might survive a conservative assumption about the amount of confounding?

Tools we will look at:

We will derive and illustrate a number of tools:

1. Two simple statistics for regression tables, conveying sensitivity of a result
2. Contour plots exploring how point estimate, t-statistic, CI would change depending on hypothesized degree of confounding
3. An approach that transforms assumptions about the strength of observables vs. confounders into bounds on the allowable amount of confounding.
4. Analysis of “extreme scenarios” when investigators are in a weak position to argue for limited confounding.

Omitted Variable Bias

The (old) idea of "omitted variable bias" (OVB):

Suppose an investigator wishes to run:

$$Y = \hat{\tau} D + \boldsymbol{X}\hat{\beta} + \hat{\gamma} Z + \hat{\epsilon}_{\text{full}} \tag{1}$$

But, Z is unobserved, so they run:

$$Y = \hat{\tau}_{\text{res}} D + \boldsymbol{X}\hat{\beta}_{\text{res}} + \hat{\epsilon}_{\text{res}} \tag{2}$$

We can ask how $\hat{\tau}$ would change if you could add Z.

Whether that full model is a wise one to run or corresponds to the quantity of interest up to you!

Omitted Variable Bias

By "partialling out", we can use the traditional OVB formulation on multivariate data:

$$\begin{aligned}\hat{\tau}_{\text{res}} &= \frac{\text{cov}(D^{\perp \boldsymbol{X}},\ Y^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})} \\ &= \frac{\text{cov}(D^{\perp \boldsymbol{X}},\ \hat{\tau} D^{\perp \boldsymbol{X}} + \hat{\gamma} Z^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})} \\ &= \hat{\tau} + \hat{\gamma}\hat{\delta} \end{aligned} \tag{3}$$

where $Y^{\perp \boldsymbol{X}}$, $D^{\perp \boldsymbol{X}}$ and $Z^{\perp \boldsymbol{X}}$ are the variables Y, D and Z after removing the components linearly explained by $\boldsymbol{X}$ and we define $\hat{\delta} := \frac{\text{cov}(D^{\perp \boldsymbol{X}}, Z^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})}$.

Note that $\hat{\delta}$ is the coefficient you would get if you could run:

$$Z = \hat{\delta} D + \boldsymbol{X}\hat{\psi} + \hat{\epsilon}_Z \tag{4}$$

OVB continued

This leaves us with:

$$\widehat{\text{bias}} = \hat{\gamma}\hat{\delta}$$

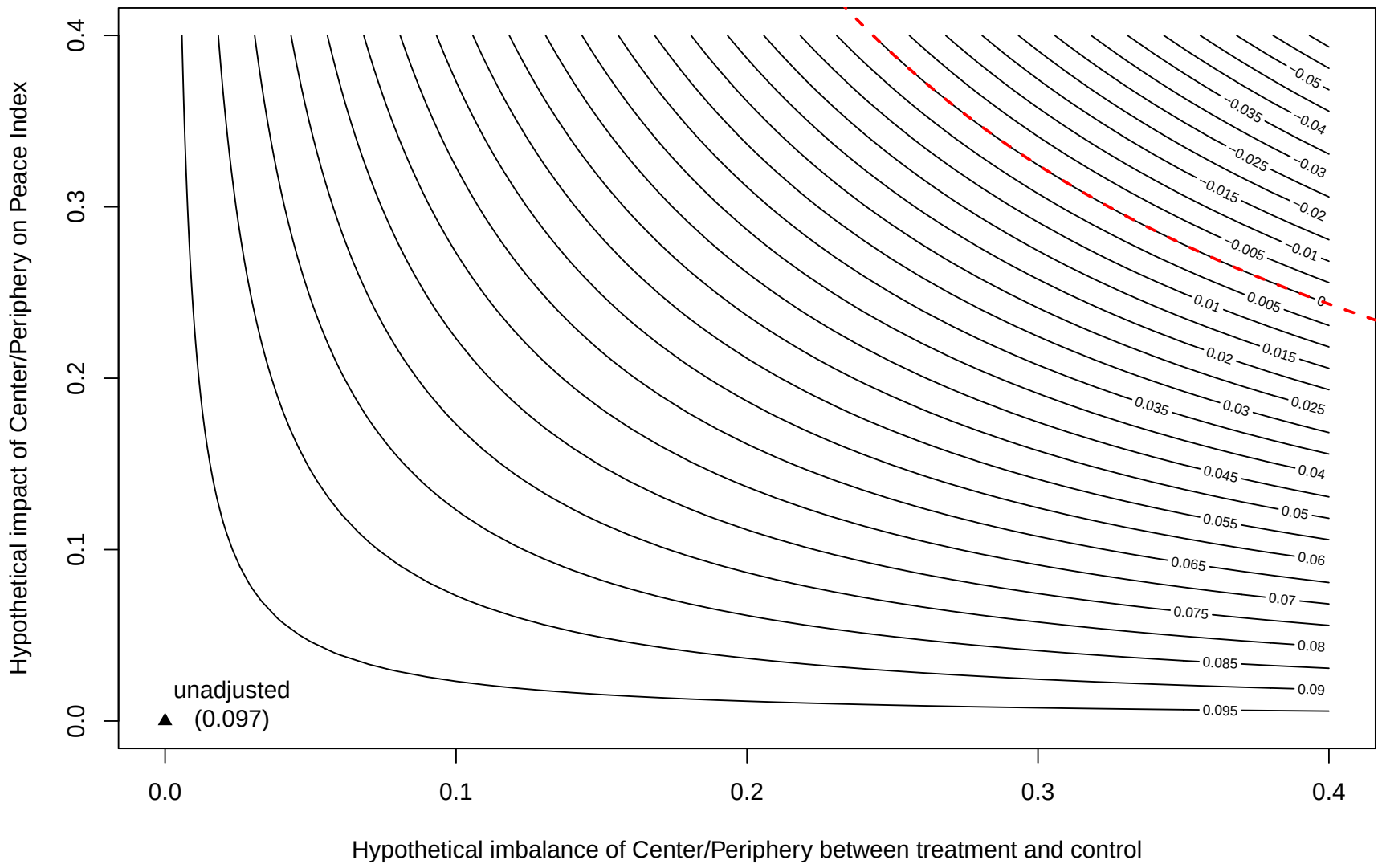
- "Bias equals impact times imbalance" (both conditional and non-causal).
- Key lesson: computing the consequence of omitting Z on the estimate of τ requires knowing *only* two things:
  - how Z relates to Y ("impact", γ)
  - how Z relates to D ("imbalance", δ)
- This gives basis for a simple sensitivity analysis without assumptions about distribution of τ, treatment assignment, etc.

Darfur example: sensitivity with OVB

Suppose a colleague argues:

> *Those in the center of the village were more likely to get hit than those on the edges, and have different attitudes.*

- How bad would this *Center* confounder have to be to be problematic?
- Consider a contour plot where, at each pair of hypothesized "impact" and "imbalance" compute adjusted effect estimate, i.e $\hat{\tau} = \hat{\tau}_{\text{res}} - \hat{\gamma}\hat{\delta}$


Hypothetical impact of Center/Periphery on Peace Index
0.0
0.1
0.2
0.3
0.4
unadjusted
(0.097)
-0.05
-0.035
-0.04
-0.025
-0.03
-0.015
-0.02
-0.005
-0.01
0.01
0.005
0
0.02
0.015
0.035
0.03
0.025
0.045
0.04
0.055
0.05
0.065
0.06
0.075
0.07
0.085
0.08
0.095
0.09
0.0
0.1
0.2
0.3
0.4
Hypothetical imbalance of Center/Periphery between treatment and control

Reparameterization

Some problems with this approach:

- *Center* was binary variable, so units of measure attached to "impact" and "imbalance" are natural; but often scaling is more arbitrary. Consider abstract confounders like "Political Attitudes".

- More generally, the "name a possible confounder" approach is too narrow: need to generalize to confounders of arbitrary scale, multiple confounders.

We reparameterize "impact" and "imbalance" using partial R^2 (of Z with D and of Z with Y)

- gives scale-free yet intuitive measure of association
- turns out to make a number of other tasks possible or easier: multiple Z, non-linear action, bounding, variance

Bias formulas

$$|\widehat{\text{bias}}| = \text{se}(\hat{\tau}_{\text{res}})\sqrt{\frac{R^2_{Y\sim Z|\boldsymbol{X},D}\, R^2_{D\sim Z|\boldsymbol{X}}}{1 - R^2_{D\sim Z|\boldsymbol{X}}}(\text{df})} \tag{5}$$

$$\text{se}(\hat{\tau}) = \text{se}(\hat{\tau}_{\text{res}})\sqrt{\frac{1 - R^2_{Y\sim Z|\boldsymbol{X},D}}{1 - R^2_{D\sim Z|\boldsymbol{X}}}\left(\frac{\text{df}}{\text{df} - 1}\right)} \tag{6}$$

Backup: what is the partial R^2

First, the partial correlation, is the correlation after partialing out, e.g. $R_{Y \sim D|X} = cor(Y^{\perp X}, D^{\perp X})$. Square that to get the partial R^2.

$R^2_{D \sim Z|X}$, for example can be thought of as the "proportion of variance explained in D by Z, after accounting for X. Specifically, we mean:

$$R^2_{D \sim Z|\boldsymbol{X}} = \frac{R^2_{D \sim \boldsymbol{X}+Z} - R^2_{D \sim \boldsymbol{X}}}{1 - R^2_{D \sim \boldsymbol{X}}} \tag{7}$$

Similarly you could say:

$$R^2_{Y \sim Z|D,\boldsymbol{X}} = \frac{R^2_{Y \sim D+\boldsymbol{X}+Z} - R^2_{Y \sim D+\boldsymbol{X}}}{1 - R^2_{Y \sim D+\boldsymbol{X}}}, \tag{8}$$

Note that the direction does not matter:
$R^2_{Z \sim D} = \text{cor}(Z, D)^2 = \text{cor}(D, Z)^2 = R^2_{D \sim Z}$.

Deriving the bias formula in the R^2 parameterization

By FWL,

$$\begin{aligned}
\widehat{\text{bias}} &= \hat{\delta}\hat{\gamma} \\
&= \left(\frac{\text{cov}(D^{\perp \boldsymbol{X}},\ Z^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})}\right)\left(\frac{\text{cov}(Y^{\perp \boldsymbol{X},D},\ Z^{\perp \boldsymbol{X},D})}{\text{var}(Z^{\perp \boldsymbol{X},D})}\right) \\
&= \left(\frac{\text{cor}(D^{\perp \boldsymbol{X}},\ Z^{\perp \boldsymbol{X}})\text{sd}(Z^{\perp X})}{\text{sd}(D^{\perp X})}\right)\left(\frac{\text{cor}(Y^{\perp X,D},\ Z^{\perp X,D})\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(Z^{\perp \boldsymbol{X},D})}\right) \\
&= \left(\frac{\text{cor}(Y^{\perp \boldsymbol{X},D},\ Z^{\perp \boldsymbol{X},D})\text{cor}(D^{\perp \boldsymbol{X}},\ Z^{\perp \boldsymbol{X}})}{\frac{\text{sd}(Z^{\perp \boldsymbol{X},D})}{\text{sd}(Z^{\perp \boldsymbol{X}})}}\right)\left(\frac{\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(D^{\perp \boldsymbol{X}})}\right) \qquad (9)
\end{aligned}$$

Noting that $\text{cor}(Y^{\perp \boldsymbol{X},D}, Z^{\perp \boldsymbol{X},D})^2 = R^2_{Y\sim Z|\boldsymbol{X},D}$, that $\text{cor}(Z^{\perp \boldsymbol{X}},\ D^{\perp \boldsymbol{X}})^2 = R^2_{D\sim Z|\boldsymbol{X}}$, and that $\frac{\text{var}(Z^{\perp \boldsymbol{X},D})}{\text{var}(Z^{\perp \boldsymbol{X}})} = 1 - R^2_{Z\sim D|\boldsymbol{X}} = 1 - R^2_{D\sim Z|\boldsymbol{X}}$, we can write 9 as simply,

$$|\widehat{\text{bias}}| = \sqrt{\frac{R^2_{Y\sim Z|D,\boldsymbol{X}}\ R^2_{D\sim Z|\boldsymbol{X}}}{1 - R^2_{D\sim Z|\boldsymbol{X}}}}\left(\frac{\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(D^{\perp \boldsymbol{X}})}\right) \qquad (10)$$

Bias anatomy

$$\text{relative bias} = \frac{\overbrace{|R_{Y\sim Z|\boldsymbol{X},D} \times f_{D\sim Z|\boldsymbol{X}}|}^{\text{bias factor}}}{\underbrace{|f_{Y\sim D|\boldsymbol{X}}|}_{\text{partial f of D with Y}}} = \frac{BF}{|f_{Y\sim D|\boldsymbol{X}}|} \tag{11}$$

- f is from the (partial) Cohen's f^2, which is $R^2/(1-R^2)$.
- Size of relative bias is the BF compared to $f_{Y\sim D|\boldsymbol{X}}$.
  - $f_{Y\sim D|\boldsymbol{X}}$ is just a transformation of $R_{Y\sim D|X}$. (Also simply $t_D/\sqrt{df}$)
  - $f_{Y\sim D|\boldsymbol{X}}$ is the *only* property of the data that enters.
- This form reminds us that sensitivity is an identification problem, unrelated to sample size.

Variance anatomy

$$\text{relative variance} = \overbrace{\left(1 - R^2_{Y \sim Z|\boldsymbol{X},D}\right)}^{\text{VRF}} \underbrace{\left(\frac{1}{1 - R^2_{D \sim Z|\boldsymbol{X}}}\right)}_{\text{VIF}} \overbrace{\left(\frac{\text{df}}{\text{df} - 1}\right)}^{\text{change in df}}$$

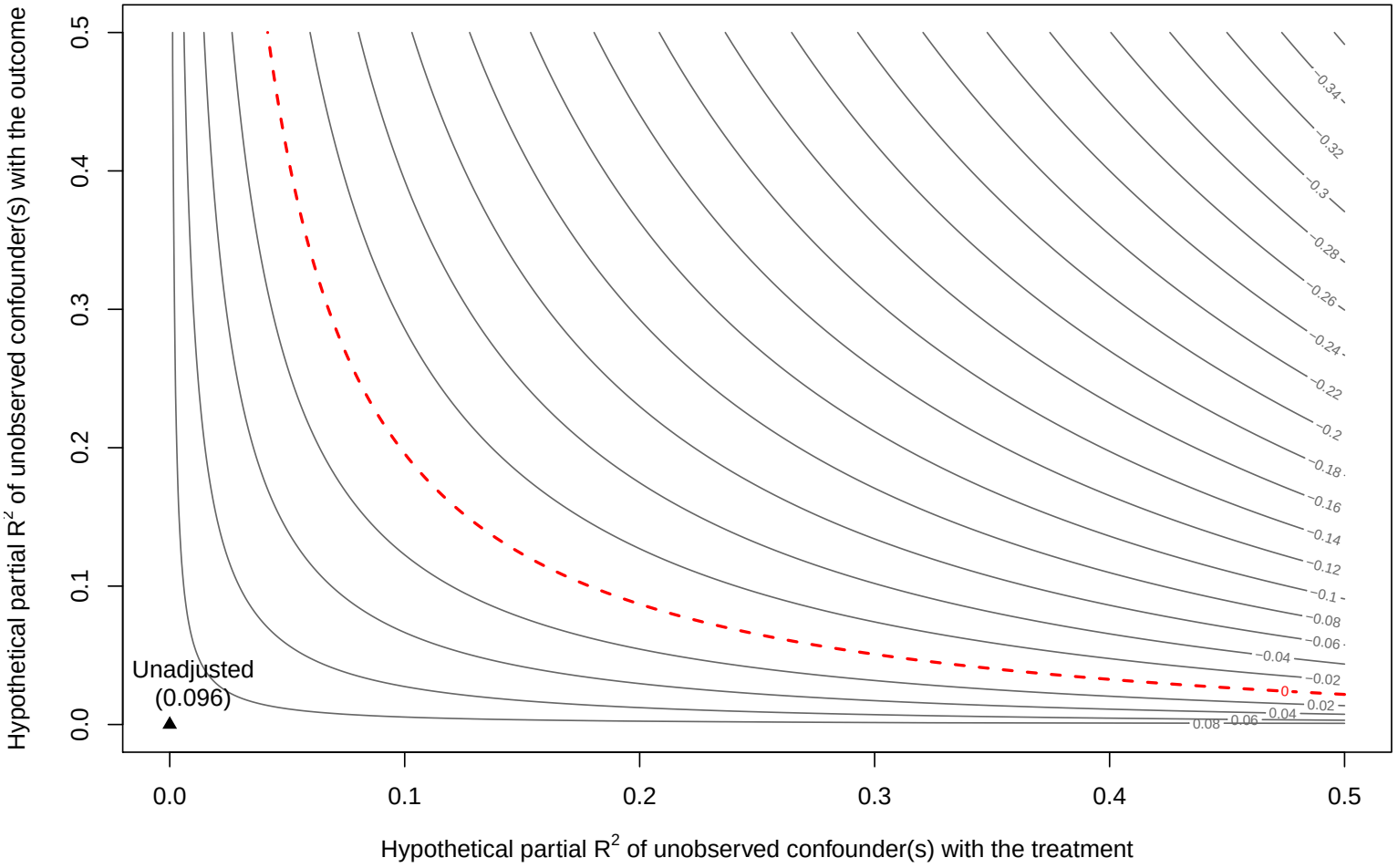
$$= \text{VRF} \times \text{VIF} \times \text{change in df} \tag{12}$$

Tools

1. Two simple statistics for routine reporting: RV and $R^2_{Y \sim D|\boldsymbol{X}}$
2. Contour plots of various confounding-adjusted quantitites, with bounds
3. Extreme-scenario plots when large proportion of residual may be due to confounder.

First, good to have this in your head:

Figure: Contour plot with partial R^2



Simple statistics for reporting sensitivity

- *Robustness Value* (RV): what is the point with equal association to treatment and outcome that would eliminate the effect?
  - RV_q: What confounder, with equal association...would bring down your estimate by some proportion q?
  - RV_α: What confounder, with equal association...would bring down your estimate so that t-statistic corresponds to some α level?
- $R^2_{Y \sim D|\boldsymbol{X}}$: partial variance explained by treatment
  - if confounder explained *all* residual variance of the outcome, to bring the estimated effect down to zero, this would equal the partial R^2 of the confounder with the treatment.

Augmented Regression Table: Darfur example

Outcome: *Peace Index*

Treatment:	Est.	SE	t-value	$R^2_{Y\sim D\mid \boldsymbol{X}}$	RV	$\text{RV}_{\alpha=0.05}$
Directly Harmed	0.097	0.023	4.18	2.2%	13.9%	7.6%
df = 783				*Bound (1 × female):* R^2_y = 12%, R^2_d = 1%		

Somebody tell us:

- what does this RV mean?
- what does thie $RV_{\alpha=0.05}$ mean?
- what does this $R^2_{Y\sim D\mid X}$ mean?

Adding bounds

How can our substantive knowledge of situation and research design enable plausibility judgements about what confounders might exist?

- A common existing approach we'll refer to as "informal benchmarking" considers a given covariate X_j and its observed relationship (e.g. partial R^2) with D and with Y as benchmarks.
- They are intended to help "calibrate our intuition" or tell us how a confounder "not unlike X_j" would change our answer.
- But, the bias caused by the confounder changes these observed statistics in problematic ways (even when $\boldsymbol{X} \perp Z$).

Confounding of informal benchmarks

Consider how the coefficient on $\boldsymbol{X}$ is altered by omissions of Z, using the same OVB approach:

$$\hat{\beta}_{\text{res}} - \hat{\beta} = \hat{\gamma}\hat{\psi} \tag{13}$$

where $\hat{\psi}$ is obtained from the regression $Z = \hat{\delta}D + \boldsymbol{X}\hat{\psi} + \hat{\epsilon}_Z$.

- Note that $\hat{\psi}$ can be non-zero even if $\boldsymbol{X} \perp Z$, because D is a collider
- Thus, claims comparing $\hat{\beta}_{\text{res}}$ with a hypothesized confounder, as if $\hat{\beta}_{\text{res}} \approx \hat{\beta}$ can be very misleading.

Illustrating the benchmark problem

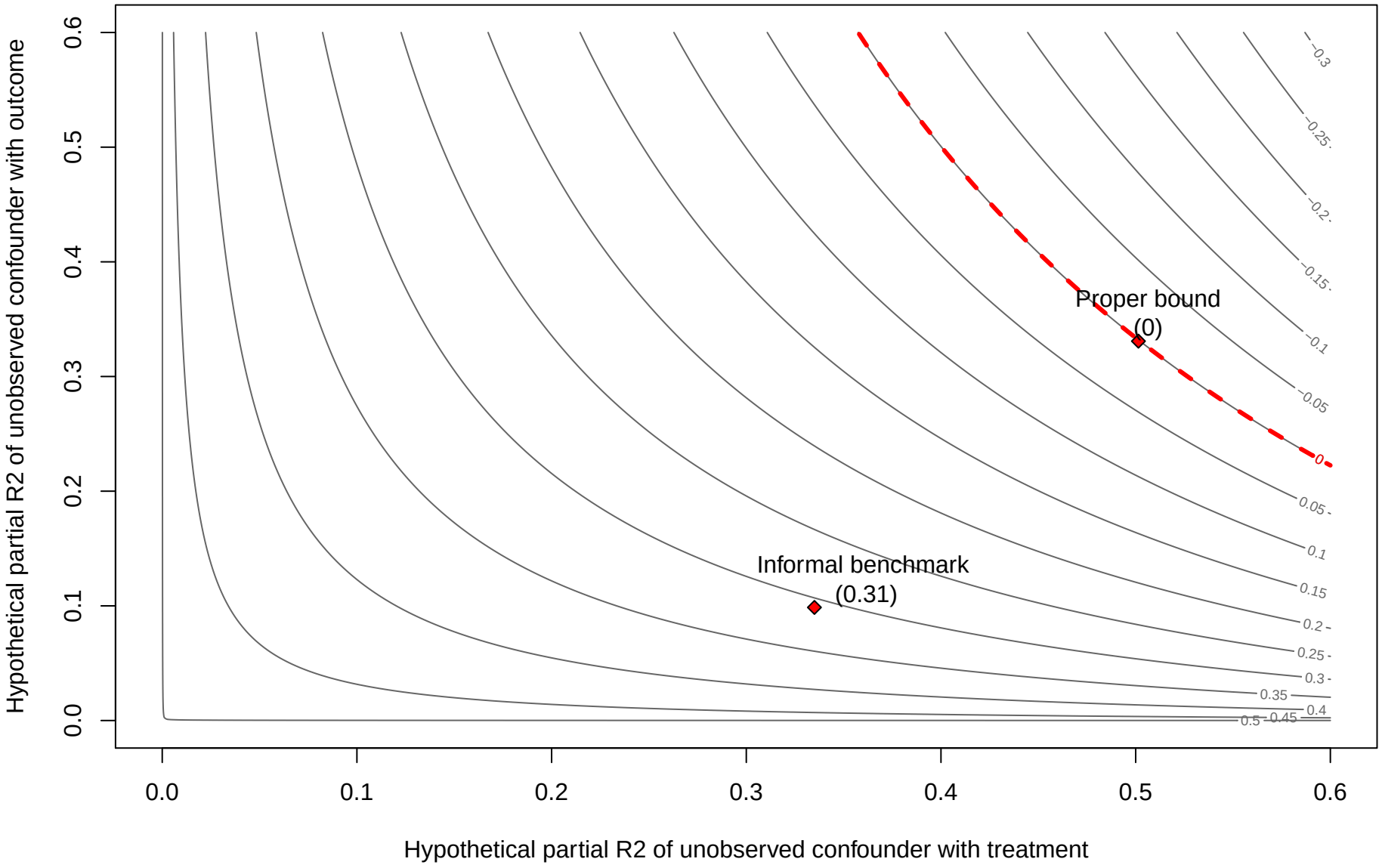
Suppose:

$$X = \epsilon_x \tag{14}$$
$$Z = \epsilon_z \tag{15}$$
$$D = X + Z + \epsilon_d \tag{16}$$
$$Y = X + Z + \epsilon_y \tag{17}$$

- No true effect of D on Y
- Z and X exactly alike in terms of relationship to D and to Y

We would thus like a benchmarking analysis to tell us that an omitted confounder "not unlike X" would indeed eliminate our estimated effect of D.

Illustrating the benchmark problem



Bounding Approach

More important is what we can actually do: Transform formal claims about how observables compare to confounders in explaining D and Y into bounds on the confounders.

- Motivation: suppose you have an observed covariate you can argue explains more of the (residual variance of) treatment assignment than confounding possibly could. Likewise for outcome variance.
- More generically: there is always some k_D, k_Y s.t.:

$$k_D := \frac{R^2_{D \sim Z | \boldsymbol{X}_{-j}}}{R^2_{D \sim X_j | \boldsymbol{X}_{-j}}}, \qquad k_Y := \frac{R^2_{Y \sim Z | \boldsymbol{X}_{-j}, D}}{R^2_{Y \sim X_j | \boldsymbol{X}_{-j}, D}} \tag{18}$$

- Can be done with any sub-group or with all covariates jointly

Result: bounding the confounding subject to assumption

Given some $\{X_j, k_D, k_Y\}$, the strength of the confounders is bounded at:

$$R^2_{D\sim Z|\boldsymbol{X}} = k_D f^2_{D\sim X_j|\boldsymbol{X}_{-j}}, \qquad R^2_{Y\sim Z|\boldsymbol{X},D} \leq \eta^2 f^2_{Y\sim X_j|\boldsymbol{X}_{-j},D} \tag{19}$$

where $\eta = \frac{\sqrt{k_Y} + \left| f_{D\sim\sqrt{k_D}X_j|\boldsymbol{X}_{-j}} \times f_{D\sim X_j|\boldsymbol{X}_{-j}} \right|}{\sqrt{1 - f^2_{D\sim\sqrt{k_D}X_j|\boldsymbol{X}_{-j}} \times f^2_{D\sim X_j|\boldsymbol{X}_{-j}}}}$.

Allows us to investigate the maximum effect a confounder at most "k times" as strong as a particular covariate X_j would have on the coefficient estimate.

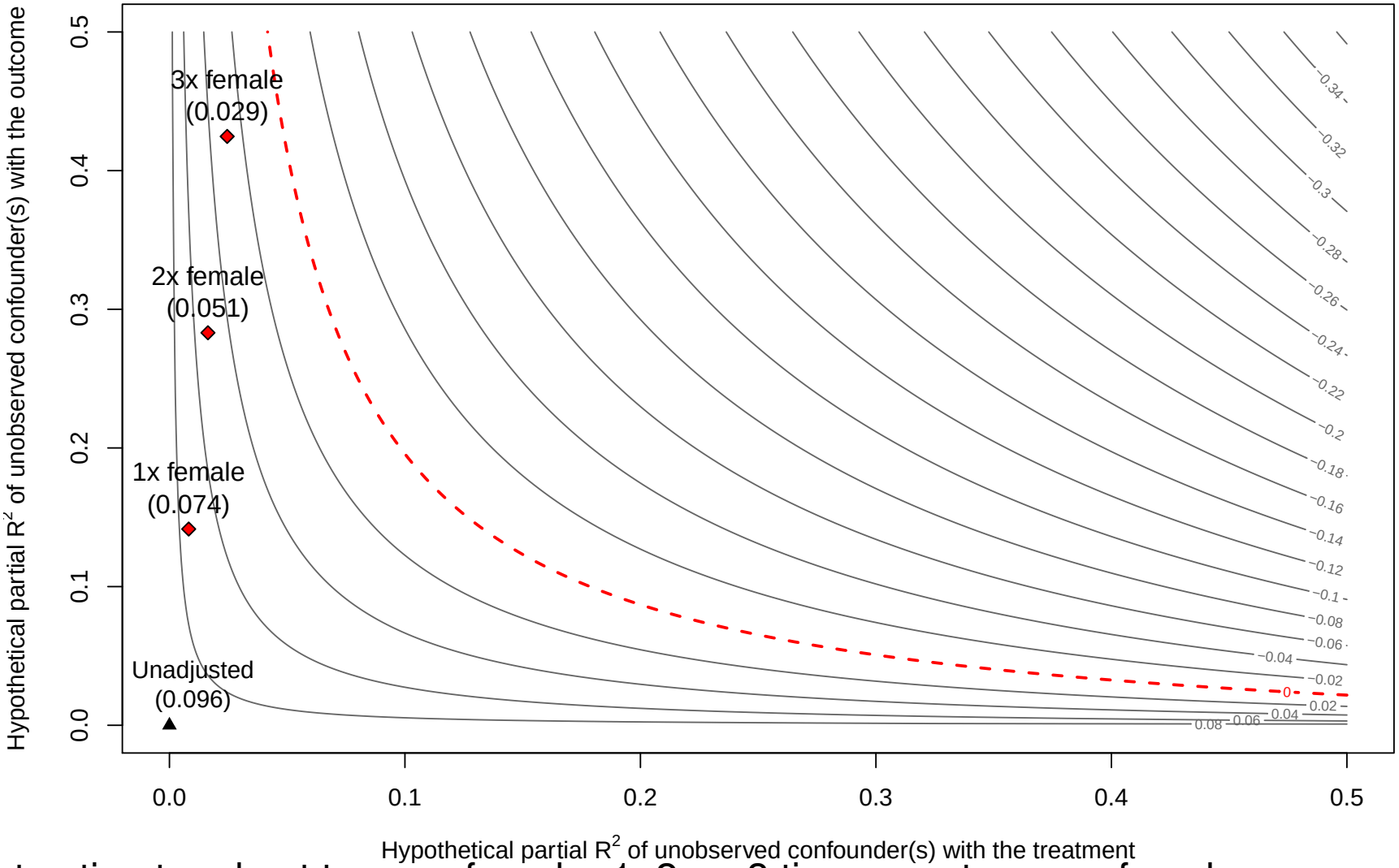
- Certain values for k_D and k_Y may be ruled out by the data (for instance, if $R^2_{D\sim X_j|\boldsymbol{X}_{-j}} = 50\%$ then k_D must be less than 1).

Confounder bounding continued

This shows the bounds implied by an assumption – you must judge/argue if its a believable assumption.

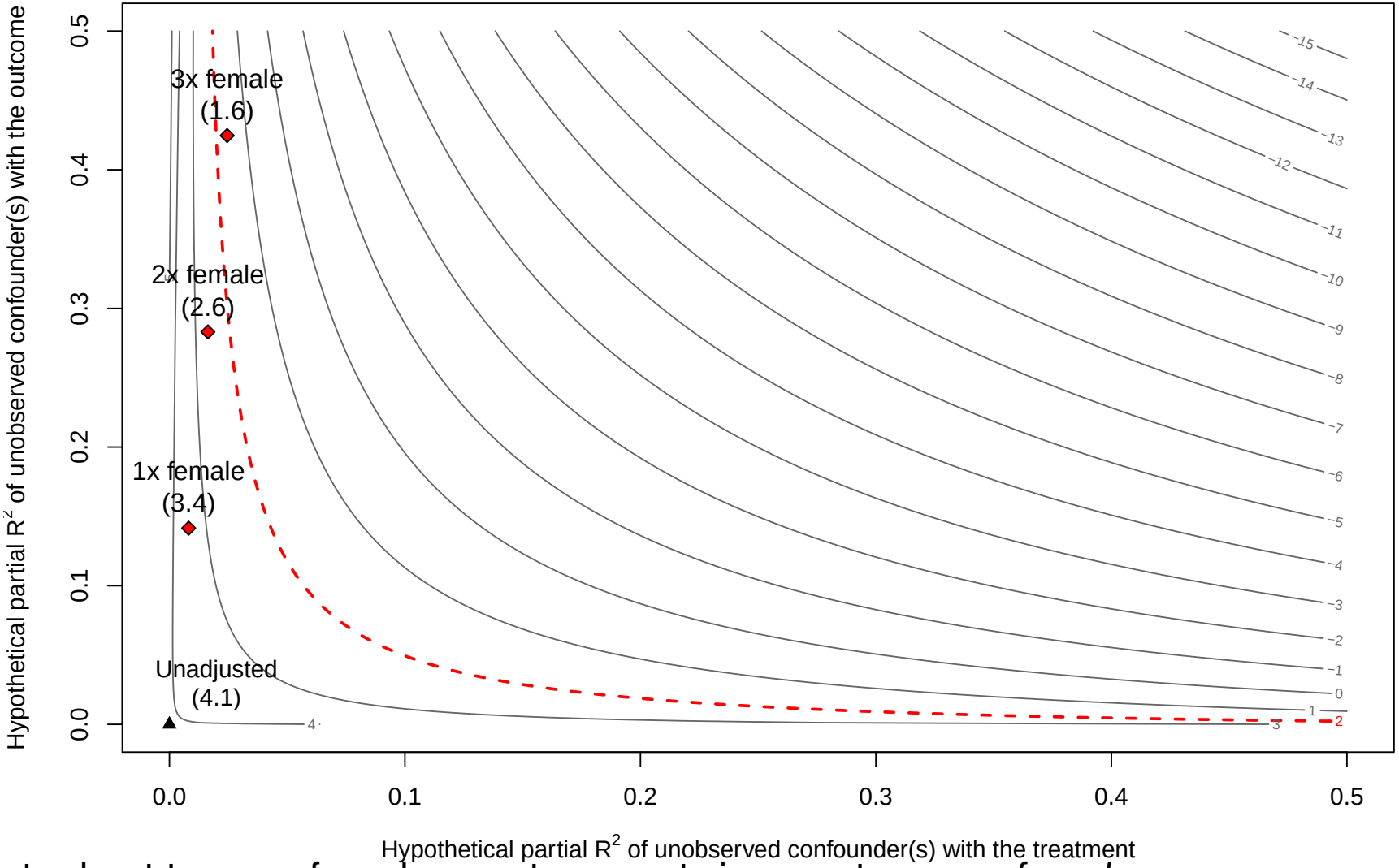
- In a poorly controlled regression, you may not be able to defend any values of k_D and k_Y for any covariate or even all covariates.
- In a natural experiment where you know about treatment assignment or the outcome, you may have more traction.
- Example: in the Darfur case, within village it is hard to imagine any basis for targeting of violence that could approach the importance of *female* indicator, so let's use it for this assumption.

Contour plots with bounds



Point estimate robust to a confounder 1, 2, or 3 times as strong as female.
(Conservative in case of multiple, non-linear confounders)

Contour plot of t-statistics



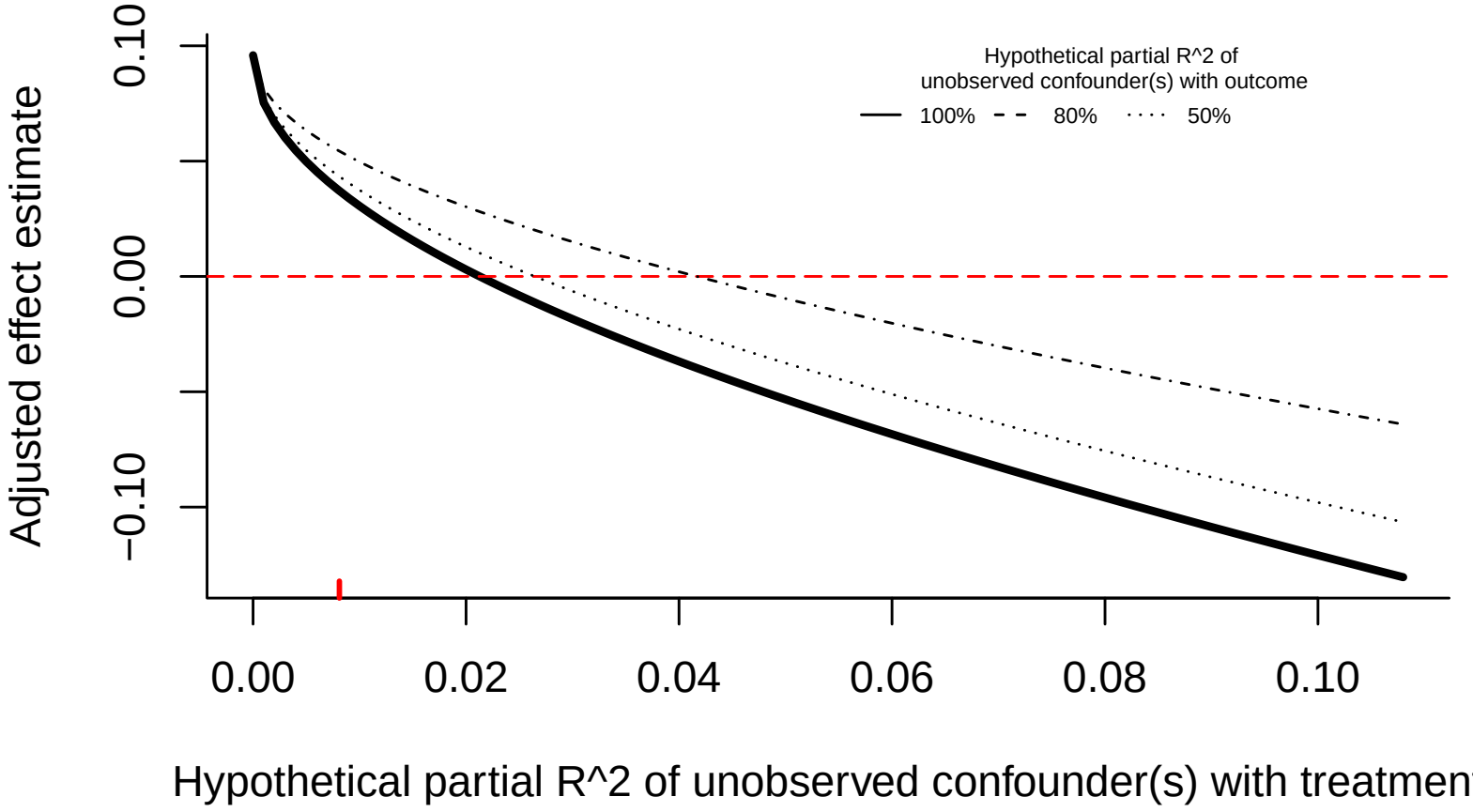
T-stat robust to a confounder as strong or twice as strong as *female*

Extreme Scenario Plots

Suppose we have cannot convincingly argue about limits to how much confounding explains of the outcome.

- Suppose all residual variation of outcome is due to Z (i.e. $R^2_{Y \sim Z|\boldsymbol{X},D} = 1$) or similar conservative assumption: then *how strongly would confounding have to be associated with treatment to be problematic?*
- Often gives bad news: you pay a price for not being able to rule out moderate confounders by design or special knowledge!

Extreme Scenario Plots: Example



In the most extreme scenario, confounder(s) have to explain 2.2% of the treatment to bring effect to zero (which is exactly $R^2_{Y \sim D|\boldsymbol{X}}$). Compare to bound obtained from "1 x female". (red marker)

Anticipating Abuses

- There cannot exist a suitable guideline for what a “good enough” RV or anything else should be. Why?
  - This is not a “test” for confounding; we are revealing what confounding would be problematic.
  - The idea is to refine debate about confounding in your situation, not to hide behind a rule.
- Readers should demand good stories for bounding exercise
  - point is not to show “it survives a bound”, but rather to convince us that confounding is not likely to be k-times worse than a given covariate.
- Extreme scenario analysis does not excuse poor design – you pay a cost for inability to rule out confounders.

Arguing (from conclusion of the software paper)

1. A causal interpretation of the research conclusion may be defended by claiming that, given the way injuries (the "treatment") occurred, the scope for targeting particular types of individuals was quite limited... aircraft dropped makeshift and unguided bombs and other objects over villages, and militia raided without concern for who they would attack—the only known major exception to this, due to sexual assaults, was targeting gender, which is also one of the most visually apparent characteristics of an individual. Thus, a confounder twice as strong as female would be surprising.

2. Similarly, for the causal conclusion to be persuasively dismissed, it does not suffice to argue that *some* confounding might exist. Helpful skepticism must articulate why a confounder that explains more than twice of the variation of the treatment assignment than the covariate female is plausible. Otherwise, the putative confounder cannot logically account for all the observed association, even if it explains all or some large portion of the residual outcome variation.

This is the kind of argumentation I expect when using benchmarks.

Good practice:

1. For any regression, provide a table with RV, $\text{RV}_{\alpha=0.05}$, and $R^2_{Y\sim D|\boldsymbol{X}}$
2. Say what you can about whether confounders that would alter conclusions are likely, given design and knowledge
   - If informative, show bounds and explain why that covariate is likely stronger than confounder
   - Show contour plots to examine more elaborate arguments
3. Show extreme scenario plots, particularly if we have few arguments with which to limit $R^2_{Y\sim Z|D,\boldsymbol{X}}$.
4. These tools also empower you as a reader or reviewers: if researchers do not reveal sensitivity of their estimates, you can do much of this with just t and df.

Bigger picture: Moving away from "is it identified", towards more informative debate over "how much confounding would change the conclusion".